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CHAPTER 26 — DEBUGGING TOOLS, TRACING & INSTRUMENTATION

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A maximum-detail, RTOS/QNX-style breakdown of how to debug MMU systems, trace page-table behavior, inspect TLB and walk-cache state, instrument hypervisors, and build reliable diagnostic tooling.

This chapter focuses on **practical debugging**, giving you the tools and techniques used by kernel engineers, hypervisor developers, and embedded firmware teams.

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26.1 — Why MMU Debugging Is Hard

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MMU bugs are notoriously difficult because:

- faults occur far from the root cause
- stale TLB entries hide descriptor changes
- walk-cache retains old table pointers
- device memory behaves differently
- SMP cores see different translations
- Stage-2 faults propagate differently
- caches obscure memory writes
- ordering issues cause nondeterministic failures

This chapter provides **real debugging tools** to make MMU issues visible.

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26.2 — Categories of MMU Debugging Tools

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You need tools in four categories:

1. Page-Table Inspection Tools

Dump and decode descriptors.

2. TLB & Walk-Cache Inspection Tools

Reveal stale entries and aliasing.

3. Fault Analysis Tools

Decode ESR/FAR/HPFAR.

4. Hypervisor Tracing Tools

Track Stage-2 behavior.

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26.3 — Page-Table Dumpers

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A page-table dumper prints:

- VA → PA mappings
- descriptor type (block/page/table)
- permissions (AP, PXN, UXN)
- attributes (AttrIndx → MAIR)
- shareability
- AF bit
- contiguous bit
- granule size
- level (L1/L2/L3)

Essential Fields to Display

```
Descriptor value (raw)
Type (block/page/table)
PA_base
```

```
AP
PXN/UXN
SH
AttrIndx
AF
Contiguous
```

Why this matters

You can instantly see:

- incorrect attributes
- incorrect permissions
- incorrect alignment
- incorrect shareability
- incorrect granule
- incorrect block/page usage

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26.4 — TLB Dumpers

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TLB dumpers show:

- VA → PA entries
- ASID
- VMID
- global/non-global flag

- permissions
- attributes
- shareability

Why this matters

TLB dumpers reveal:

- stale entries
- aliasing
- incorrect ASID reuse
- incorrect VMID reuse
- missing TLBI

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26.5 — Walk-Cache Dumpers

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Walk-cache stores intermediate page-table entries.

Dumping walk-cache reveals:

- stale table pointers
- stale block descriptors
- stale page descriptors
- incorrect BBM sequences

Why this matters

Walk-cache bugs cause:

- silent corruption
- nondeterministic faults
- SMP inconsistencies

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26.6 — ESR/FAR/HPFAR Fault Decoders

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Fault decoders interpret:

- **ESR_ELx** — exception class + fault status
- **FAR_ELx** — faulting VA
- **HPFAR_EL2** — faulting IPA (Stage-2)

Essential ESR Fields

```
EC   (Exception Class)
DFSC (Data Fault Status Code)
IFSC (Instruction Fault Status Code)
```

Common Faults

- translation fault
- access flag fault
- permission fault
- execute fault
- alignment fault
- Stage-2 fault

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26.7 — Stage-2 Fault Tracers

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Hypervisors need tools to trace:

- IPA → PA translation
- Stage-2 descriptor selection
- S2AP/S2PXN/S2UXN enforcement
- Stage-2 walk behavior
- Stage-2 TLB entries

Why this matters

Stage-2 faults are often misdiagnosed as Stage-1 faults.

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26.8 — Descriptor Change Tracers

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Track:

- descriptor writes
- descriptor alignment
- descriptor type changes
- BBM sequences
- TLBI sequences

Why this matters

Most MMU bugs come from:

- incorrect descriptor updates
- missing barriers
- missing TLBI

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26.9 — Barrier Tracers

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Trace:

- DSB SY
- DMB SY
- ISB

Why this matters

Missing barriers cause:

- stale TLB entries
- stale walk-cache entries
- stale device writes
- nondeterministic behavior

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26.10 — Cache Maintenance Tracers

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Trace:

- DC CVAC
- DC CIVAC
- DC IVAC
- IC IVAU
- IC IALLU

Why this matters

Incorrect cache maintenance causes:

- DMA corruption
- stale descriptors
- stale code
- stale device state

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26.11 — SMP Coherency Tracers

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Trace:

- broadcast TLBI
- broadcast cache maintenance
- shareability mismatches
- per-core descriptor visibility

Why this matters

SMP bugs often appear only under load.

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26.12 — Hypervisor Tracing Tools

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Hypervisors require tools to trace:

- VMID allocation
- Stage-2 updates
- Stage-2 faults
- nested virtualization behavior
- shadow page-table updates
- device virtualization traps

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26.13 — Device Access Tracers

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Trace:

- MMIO reads
- MMIO writes
- ordering (DSB SY)
- device register access patterns

Why this matters

Incorrect MMIO mapping → device malfunction.

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26.14 — DMA Tracers

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Trace:

- DMA read/write operations
- cache maintenance
- IOMMU mappings
- DMA faults

Why this matters

DMA bugs cause silent memory corruption.

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26.15 — Interrupt Tracers

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Trace:

- interrupt routing
- interrupt masking
- interrupt injection
- GIC register access
- virtual interrupt behavior

Why this matters

Interrupt/MMU bugs cause nondeterministic crashes.

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26.16 — Page-Fault Logging

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Log:

- VA
- IPA (if Stage-2)
- PA (if resolved)
- ESR
- FAR
- HPFAR
- ASID
- VMID
- descriptor values

Why this matters

Page-fault logs reveal:

- missing mappings
- incorrect permissions
- incorrect attributes
- incorrect Stage-2 behavior

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26.17 — Memory Layout Visualization Tools

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Visualize:

- VA space
- IPA space
- PA space
- block/page boundaries
- MMIO regions
- kernel/user split
- Stage-2 overlays

Why this matters

Visual tools reveal fragmentation and misalignment.

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26.18 — Common Debugging Mistakes

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Mistake 1 — Not dumping raw descriptors

You miss bit-level errors.

Mistake 2 — Not dumping TLB

You miss stale entries.

Mistake 3 — Not dumping walk-cache

You miss BBM errors.

Mistake 4 — Not decoding ESR/FAR

You misdiagnose faults.

Mistake 5 — Not tracing barriers

You miss ordering bugs.

Mistake 6 — Not tracing MMIO

You miss device ordering bugs.

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26.19 — Full Debugging Example

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Symptom

Random crashes under SMP load.

Tools Used

- TLB dumper
- walk-cache dumper
- descriptor tracer
- barrier tracer

Findings

- stale L2 block descriptor in walk-cache
- missing DSB SY after descriptor update
- missing TLBI

Fix

```
STR x0, [descriptor]
DSB SY
TLBI VAE1, x1
DSB SY
ISB
```

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26.20 — Chapter Summary

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This chapter covered:

- page-table dumpers
- TLB dumpers
- walk-cache dumpers
- ESR/FAR/HPFAR decoders
- Stage-2 tracers
- descriptor tracers
- barrier tracers
- cache maintenance tracers
- SMP coherency tracers
- hypervisor tracing tools
- device access tracers
- DMA tracers
- interrupt tracers
- page-fault logging
- memory visualization tools
- common debugging mistakes
- full examples